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Comparative Performance Analysis of TMD based Multi-Bridge Channel Field Effect Transistor (MBCFET)

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Abstract—Shifting from planer FET to FinFET has allowed us to significantly reduce the size of transistors while maintaining high-performance characteristics. However, further area reduction has again driven us to technological limits. To enhance the performance of next-generation devices, Nanowire FET designs show some desirable characteristics. Furthermore, to enhance the control limitations, a slight variation of the Nanowire FET, known as the Multi-Bridge Channel FET (MBCFET) is introduced. In this research, we aim to provide an optimized design of MBCFET. Initially, an analytical study of MBCFET concerning Threshold Voltage, Subthreshold Swing (SS), and I_{on}/I_{off} ratio by varying different device parameters is carried out. From the analytical study, we proposed an optimized MBCFET which exhibits superior performance compared to conventional MBCFET devices. Our optimized MBCFET fabricated with a Transition Metal Dichalcogenide (TMD): Tungsten Ditelluride (WTe₂) nanosheet and Niobium pentoxide (Nb₂O₅) as high-k dielectric shows a Subthreshold Swing (SS) of 64.60mV/dec and a Threshold voltage of 0.57V at room temperature. A comparative analysis of the results shows that it provides a 2.53 times better I_{on}/I_{off} ratio than the conventional devices. Finally, a comparative study of MBCFET with existing FinFET and Nanowire FET designs with nearly identical parameters is also conducted. Therefore, MBCFET's superior performance will make it a desirable choice for Fabrication Industries to implement it in their power-efficient devices.

Keywords—MBCFET, Threshold Voltage, Subthreshold Swing, I_{on}/I_{off} Ratio, TMD, Tungsten Ditelluride, Niobium pentoxide

I. INTRODUCTION

With the increasing demands for higher performance, developers and researchers are constantly scaling down the size of transistors to pack more transistors in a more compact and power-efficient package. But continuous downscaling of transistor channel length results in several types of technological limitations like Short Channel Effects (SCEs), Source-Drain-Tunnelling (SDT), and more which causes severe performance degradation. To overcome these obstacles, diverse structures, and the design of FETs are thoroughly investigated. In recent years, three-dimensional (3-D) transistor structures like FinFET, Nanowire FET, and MBCFET are highly proposed and extensively studied [1]-[4]. The multi-gate model of these devices offers substantially noticeable advantages by enhancing the I_{on}/I_{off} ratio,

reducing the sub-threshold and tunnelling leakage currents as compared to the standard FET structures [5]-[8].

Moreover, semiconductor materials also heavily influence the characteristics of a device. For many years, silicon type materials were at the centre of focus of the most device research. But as aggressive downscaling of devices continue, silicon has already reached its performance limits. An evanescent spark of hope was created for graphene-related researches, but soon it became clear that due to its gapless nature, it is not an appropriate material for FETs [9]-[10]. For having a zero bandgap, it restricts the device to switch off properly, which downgrades the device performance. To solve this problem, researchers turn to Transition Metal Dichalcogenides (TMD) as a prospective alternative because of its numerous advantages, such as high magnetoresistance and electrical superconductivity [11]. In past years, extensive research has been going over FET devices using TMD to further improve its performance and reduce leakage. Yang L et al. has proposed a device with a channel length of 10 nm which has an average subthreshold swing of 200 mV/dec and an I_{on}/I_{off} ratio greater than [12]. In addition, Xu K et al. have proposed a device with a channel length of 6 nm. The minimum subthreshold swing of his device was around 140 mV/dec and an I_{on}/I_{off} ratio is near [13].

In this study, we conducted a detail numerical analysis on MBCFET by varying device material, dielectric material and thickness of the oxide. Based on this study, we handpicked the best performing and compatible parameters and presented an optimized MBCFET device. Furthermore, we have taken an approach towards a comparative study among three-dimensional FinFET, Nanowire FET and an MBCFET device by varying channel length and temperature.

II. DEVICE STRUCTURE

In FinFET designs, the gate oxide surrounds three sides of the structure. This improves control over the channel. However, further downscaling causes decrease in overall control. To improve control, Nanowire FETs were introduced, where the channel is surrounded by gate oxide in all four sides. However, Nanowire FET displays poor control as the effective channel moves further away from the gate oxide and semiconductor interface. Therefore, an improved Gate-All-Around FET (GAAFET) known as the Multi-Bridge Channel Field Effect Transistor (MBCFET), has displayed better

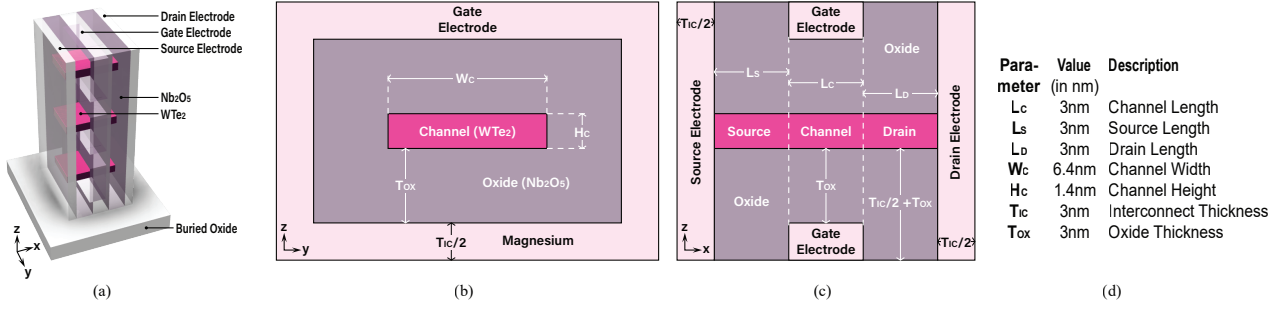


Fig. 1. (a) 3D model of the Device (b) Cross-section at the centre along drain-source (c) Cross-section at the centre from side (d) Parameter Description

performance than all other existing FET designs as they are fabricated with nanosheets instead of nanowires. MBCFET structure provides a higher level of scalability over FinFET designs as the channel width can be adjusted to the specific requirements. Moreover, the nanosheets are vertically stacked instead of horizontal paralleling, resulting in a reduction of the area required.

In this study, we designed an MBCFET with WTe_2 , a Transition Metal Dichalcogenide (TMD) nanosheet material. TMD materials show graphene like high conductivity while retaining the characteristics of a semiconductor. We also designed a Nanowire FET, a FinFET and an MBCFET with the identical material, doping, temperature, biasing, with equal cross-section area and channel length to conduct a performance comparative study.

The MBCFET model is fabricated over a buried oxide layer made of SiO_2 on a silicon substrate. The device is a three terminal device as shown in Fig. 1. (a). The default semiconductor material is a WTe_2 monolayer. The source and drain structures are identical having donor concentration of $5e+19 \text{ cm}^{-3}$. The channel is lightly doped with acceptor concentration of $1e+15 \text{ cm}^{-3}$. The semiconductor material is in Ohmic contact with source and drain electrodes made of Magnesium. The high- κ Niobium pentoxide (Nb_2O_5) insulation covers the remaining sides with a thickness of T_{ox} and the gate electrode surrounds the gate oxide with a thickness of $T_{ic}/2$ as in Fig. 1. (b). The Fig. 1. (b) and (c) show the cross-section at the centre along YZ and ZX planes. The structure can be made wider or stacked vertically as per design and performance requirements.

Device Material, oxide material, oxide thickness and channel length are varied for comparative analysis. The remaining geometric parameters are the same as described in Fig. 1. (d).

III. DEVICE CHARACTERISTIC

The threshold voltage of a device is a critical performance criterion. The threshold voltage of an FET can be noted as the minimum gate to source voltage (V_{gs}), which is needed to turn on the device. Any change in the device structure or material properties can lower down the threshold voltage resulting in a substantial change in leakage power consumption. So, to have a reliable threshold voltage, an optimized device design and material choice is needed. The V_{th} of an FET device can be expressed as (1) [14],

$$V_{th} = V_{FB} - 2\phi_p - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} \quad (1)$$

Here, V_{FB} represents the flat band voltage of the device, which indicates the difference in work functions between the metal and the semiconductor. $2\phi_p$ indicates the gate voltage that is necessary to change the surface potential of the device. Again Q_{B0}/C_{ox} is the gate voltage component necessary to offset the depletion region charge, and Q_{ox}/C_{ox} is the gate voltage that usually offsets the fixed charges at the gate oxide and channel material-oxide junction.

The subthreshold swing indicates the change of gate to source voltage required in order to have a decade change in the drain-source current. It can be calculated from the inverse slope of the log of drain current vs. gate voltage in the subthreshold region. Normally subthreshold swing is desired to be smaller to have a faster transition between off current (I_{off}) and on current (I_{on}). Studies show that a device having lower subthreshold swing tends to have lower leakage current consumption and higher threshold voltage which increases the performance of the device to a great extent. According to the Boltzmann approximation, the ideal subthreshold swing is considered to be around 60 mV/Dec. However, it has been observed that even lower subthreshold swing can be achieved by using materials with a sharply decreasing Density of States (DOS) [15]. The subthreshold swing for an FET device can be expressed as (2) [16],

$$SS = \frac{dV_{gs}}{d(\log I_{ds})} \quad (2)$$

Here, V_{gs} indicates the gate voltage and I_{ds} is the drain current. The I_{on}/I_{off} ratio can portray the overall performance of a device. It is basically the current ratio of the device at $V_{gs} = V_{DD}$ is to $V_{gs} = 0$. The I_{off} current typically indicates the leakage current of an FET when zero bias is applied to the gate while the I_{on} current is the drain to source current when a respective potential difference is applied between the gate and source of the transistor. A low amount of leakage current is desired to reduce static power loss. On the other hand, a higher value of on current reduces the gate switching delay. So, a high I_{on}/I_{off} ratio indicates the improvement in overall performance.

IV. OPTIMIZATION OF MBCFET

Initially, comparative analysis of the MBCFET device in terms of varying parameters such as channel length, device

material, oxide material and thickness are carried out using numerical device simulation tools- TCAD. We have presented the influence of these parameters on Threshold Voltage, Subthreshold Swing and I_{on}/I_{off} ratio.

A. Effect of Device Material

After reducing the dimension of any material down to a certain extent, structure integrity becomes critical. However, WTe_2 can be formed down to single molecular layer thickness. It exhibits higher carrier mobility, which tend to have greater conductivity. This naturally allows us to have a higher I_{on}/I_{off} ratio and lower subthreshold swing. Furthermore, materials with higher atomic packing density tend to show higher capacitance which results in a lower threshold voltage. Fig. 2. demonstrates the effect of different materials for MBCFET fabrication. Due to the mobility factor and the monolayer thickness, WTe_2 is showing dominating results compared to the Ge and Si under the same circumstances. Therefore, we will conduct the rest of our studies using WTe_2 as our device material.

B. Effect of Dielectric Material

Scaling down the gate size increases the capacitive value between the semiconductor and dielectric materials, which drives more current, ultimately improving the performance. But, after scaling down to a certain level, the dielectric breakdown becomes eminent, limiting the device performance. Using high- κ materials allows us to increase in gate capacitance without actually reducing the oxide thickness. The increase in gate capacitance reduces the subthreshold swing and increases threshold voltage, which implies to better control over the device (shown in Fig. 3). Increment in capacitance also increases the conductivity of the channel which leads to a higher I_{on}/I_{off} ratio. Therefore, we used material with a dielectric constant of 35, which is Niobium pentoxide. We are using Niobium Pentoxide ($\kappa=35$) over Titanium dioxide ($\kappa=95$) because there is not much significant change in I_{on}/I_{off} ratio and Titanium dioxide is much more brittle.

C. Effect of Oxide Thickness

With the reduction of gate oxide thickness, the gate channel capacitance increases proportionally. As the channel

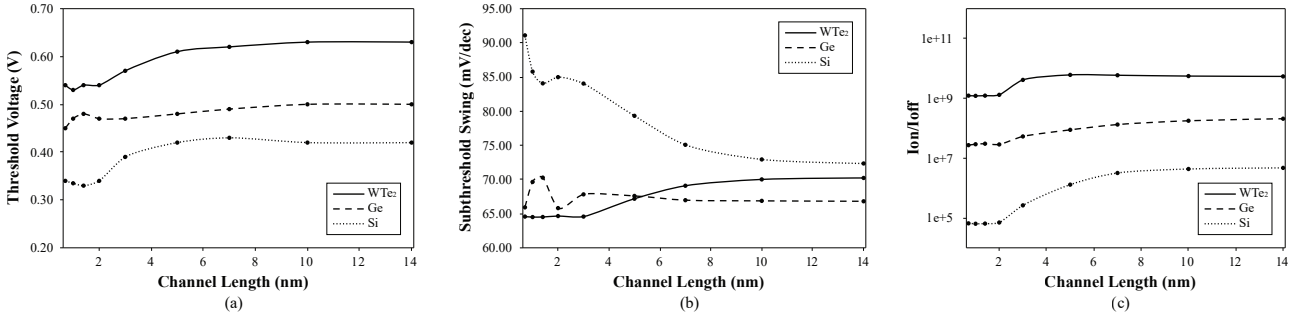


Fig. 2. Effect of Device Material on (a) Threshold Voltage (b) Subthreshold Swing (c) I_{on}/I_{off} ratio

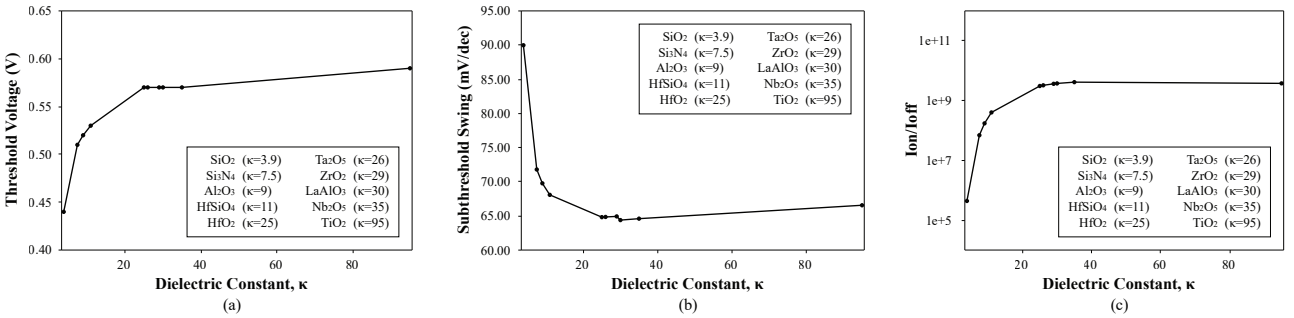


Fig. 3. Effect of Dielectric Material on (a) Threshold Voltage (b) Subthreshold Swing (c) I_{on}/I_{off} ratio

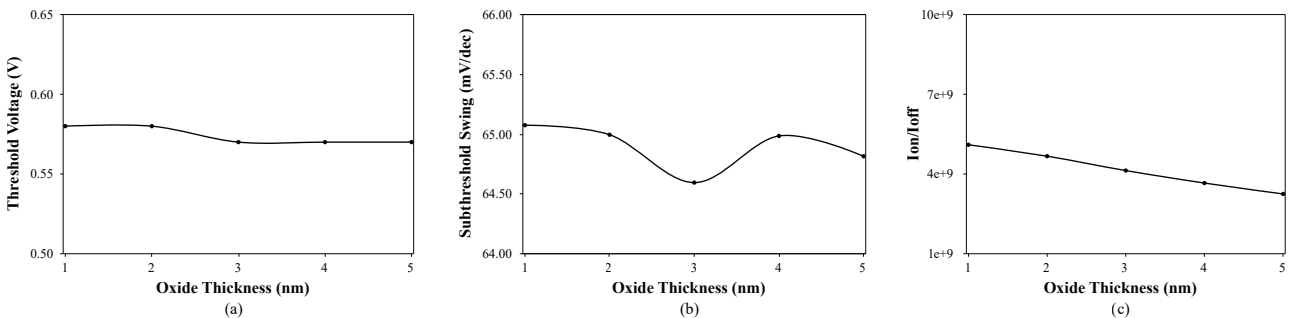


Fig. 4. Effect of Oxide Thickness on (a) Threshold Voltage (b) Subthreshold Swing (c) I_{on}/I_{off} ratio

capacitance increases, more carriers are accumulated in the channel region, causing the channel to be formed at lower potential difference. However, the dielectric breakdown causes a gate current, which causes the threshold voltage to marginally increase. The higher conductivity causes the I_{on}/I_{off} ratio to increase also. These effects are represented in Fig. 4. Increasing the Oxide thickness causes the I_{on}/I_{off} ratio to decrease and threshold voltage to rise as described. We shall continue with 3nm oxide thickness; as current technology allows us to fabricate at 3nm technological node.

V. RESULT OF MBCFET OPTIMIZATION

For our optimized MBCFET device, we are using Tungsten Ditelluride as device material and Niobium pentoxide as the insulator material. To provide clear numerical proof of performance enhancement, we also designed a conventional MBCFET device with silicon as device material and silicon dioxide as insulator. In both cases, the oxide thickness is kept being 3nm. TABLE I. represents that, MBCFET fabricated with the optimized parameters show significant performance improvement over the conventional device.

TABLE I. PREFORMANCE COMPARISON

Parameter	Conventional	Optimized
Threshold Voltage	0.49V	0.57V
Subthreshold Swing	73.02mV	64.60mV
I_{on}/I_{off} ratio	2.81e+7	4.13e+9

The threshold voltage of our optimized device is 0.57V which is 16.23% higher than the conventional MBCFET. Moreover, our optimized MBCFET provides 11.53% better subthreshold swing over conventional device. Finally, the low

I_{off} of our optimized MBCFET results in a I_{on}/I_{off} ratio which is 147 times higher than the conventional device.

VI. COMPARISON WITH OTHER FETs

After concluding the optimization process of our proposed MBCFET device, we have also conducted a comparative study among the MBCFET, FinFET and Nanowire FET. We designed the Nanowire FET and the FinFET with parameters identical to the optimized MBCFET discussed in the previous chapter (shown in TABLE II).

TABLE II. DESIGN PARAMETER OF THE FETs

Parameter	Nanowire FET	FinFET	MBCFET
Device Material	WTe ₂	WTe ₂	WTe ₂
Dielectric Material	Nb ₂ O ₅	Nb ₂ O ₅	Nb ₂ O ₅
Doping Concentration [cm ⁻³]	5e+19	5e+19	5e+19
Background Doping [cm ⁻³]	1e+15	1e+15	1e+15
Temperature	298K	298K	298K
Drain to Source Voltage	0.70V	0.70V	0.70V
Channel Length	3.0 nm	3.0 nm	3.0 nm
Channel Width	3.0 nm	1.4 nm	6.4 nm
Channel Height	3.0 nm	6.4 nm	1.4 nm

A. Effect of Channel length

Fig. 5. indicates the effect of channel length on the performance of FET devices. As the size of devices is reduced, the electric field between the drain and source becomes stronger and causes a reduction of the threshold voltage due to

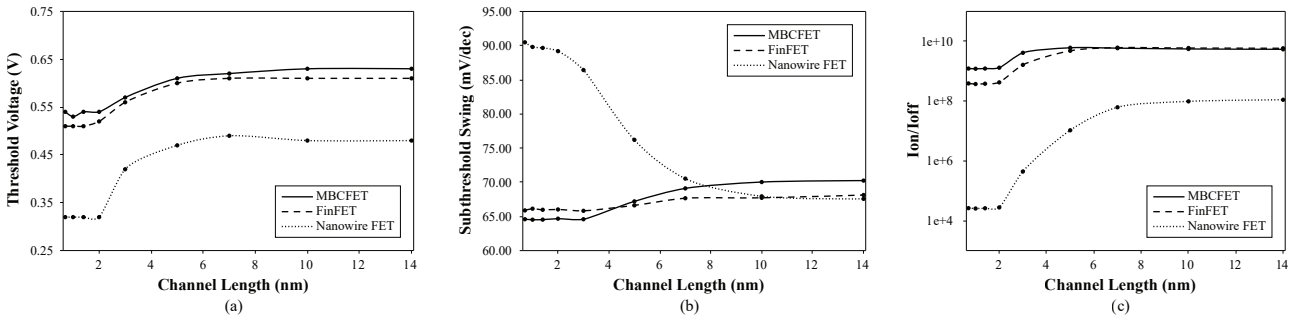


Fig. 5. Effect of Channel Length on (a) Threshold Voltage (b) Subthreshold Swing (c) I_{on}/I_{off} ratio

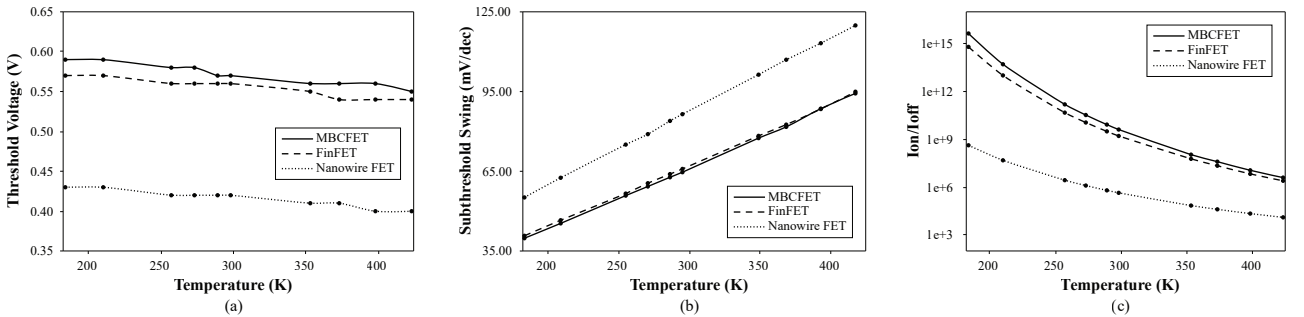


Fig. 6. Effect of Temperature on (a) Threshold Voltage (b) Subthreshold Swing (c) I_{on}/I_{off} ratio

Drain Induced Barrier Lowering (DIBL) effect. Both I_{on} and I_{off} increases, causing the I_{on}/I_{off} ratio to become smaller. The control of gate over the majority portion of the current is reduced causing Subthreshold Swing to become higher as shown in Fig. 5. (b). As MBCFET has the maximum gate surface to channel thickness ratio, inevitably causing its performance to be outstanding.

B. Effect of Temperature

The effects of temperature on the FET devices are confirmed in Fig. 6. The rising temperature causes the source-channel barrier to lower down and the number of carriers transferred from source to channel increases, which results in both I_{on} and I_{off} to increase. Therefore, with the rise of temperature, the subthreshold swing of a device starts rising as shown in Fig. 6. (b). The also I_{on}/I_{off} ratio falls, which takes a heavy toll on the device performance. Generating more carriers also lowers the threshold voltage. Even at these harsh conditions, the MBCFET device is showing promising results as it's subthreshold swing is comparatively lower than the other two devices.

The comparative analysis amongst these FET devices clearly proves MBCFET stands superior to the existing FETs in performance. The numerical conclusions of the comparative study are shown in TABLE III. The MBCFET has an excellent I_{on}/I_{off} ratio as well as it shows the potentials reach to the ideal Subthreshold Swing for respective materials.

TABLE III. PERFORMANCE COMPARISON

Parameter	Nanowire FET	FinFET	MBCFET
Threshold Voltage	0.42V	0.56V	0.57V
Subthreshold Swing	86.43mV	65.83mV	64.60mV
I_{on}/I_{off} ratio	4.44e+5	1.63e+9	4.13e+9
Performance Degradation for 80°C	14.14%	15.56%	16.37%

VII. CONCLUSIONS

Optimization and performance enhancement of MBCFET by conducting numerical analysis is carried out using computational simulations. Furthermore, a comparative study among MBCFET with existing FinFET and Nanowire FET in terms of Threshold Voltage, Subthreshold Swing and I_{on}/I_{off} ratio is presented here. We confirmed the performance superiority of MBCFET design over FinFET and Nanowire FET. The electrostatic performance of MBCFET shows very promising results in terms of leakage current and I_{on}/I_{off} ratio. With further research and tuning, it is expected that MBCFET will dominate in the fabrication industry design process. As the devices continue to scale down, MBCFET will stand as one of the excellent choices for engineers to create superior performance and power-efficient devices.

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